

Moth 2270 Review for Quiz 4

① Explain why the row space of $C(A^T)$ and the nullspaces $N(A)$ are subspaces of $\mathbb{R}^n$ and why the column space $C(A)$ and the left nullspace $N(A^T)$ are subspaces of $\mathbb{R}^m$
(Here $A \in M_{m,n}(\mathbb{R})$)

② If $A \in M_{m,n}(\mathbb{R})$ has rank r , compute the dimensions of $N(A)$, $N(A^T)$, $C(A)$, $C(A^T)$

③ Compute these spaces for $A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix}$

④ Define orthogonal vectors & show that $\underline{u}$ & $\underline{v}$ are orthogonal if and only if $\|\underline{u}\|^2 + \|\underline{v}\|^2 = \|\underline{u} - \underline{v}\|^2$

⑤ Define orthogonal spaces and show that the row & nullspaces of $A \in M_{m,n}$ are orthogonal in $\mathbb{R}^n$
(also show that the column and left nullspace are orthogonal in $\mathbb{R}^m$)

⑥ Explain why V can not be orthogonal to W in $\mathbb{R}^n$ if $\dim V + \dim W > n$.

⑦ Define orthogonal complement (of $V \subset \mathbb{R}^n$) and show that $N(A)$ is the orthogonal complement to $C(A^T)$ while $N(A^T)$ is the orthogonal complement to $C(A)$.

(use ex 5)